Metabolic-Prior Replicator Dynamics Predict Oral Microbiome Community Composition under Leave-One-Out Cross-Validation

Internal analysis document — NIFE/IKM collaboration, May 11, 2026

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Abstract

We investigate whether metabolic network constraints derived from a literature-curated and flux-balance-analysis (FBA)-augmented sign prior can improve out-of-sample prediction of oral microbiome community dynamics. A Hamilton replicator ODE with symmetric interaction matrix $\mathbf{A}$ and a three-layer sign prior — eHOMD-based Szafrański (L1) experimental metabolite flows, supplemented by AGORA2 (L2+L3) pFBA cross-feeding predictions — is fit to 16S rRNA amplicon data from 10 patients during caries recovery [Dieckow et al., 2024], aggregated to 10 class-level guilds over three weekly timepoints. Leave-one-patient-out (LOO) cross-validation on 10 folds yields: LOO-RMSE $= 0.0504 \pm 0.021$ (AGORA W$= 1.0$) vs. 0.0516 ± 0.021 (L1+L2 only) and 0.0588 (unconstrained gLV baseline); LOO Bray-Curtis (BC) $= 0.147$, 48% below the persistence null model (0.281). Full sign agreement (100%, 70/70 constrained pairs) is achieved at AGORA weight $w = 1.0$, which acts as a phase transition: below this threshold SA is 94%, above it over-constraining reduces the benefit. LOO stability analysis shows all 45 off-diagonal pairs achieve sign consistency ≥ 0.70 across folds (13 pairs at 1.00). PCA and LDA ordination of predicted versus observed week-2/3 compositions confirm that the model correctly reproduces inter-patient community structure, not just mean trajectories. Patient-specific growth vectors $\hat{b}_p$ cluster by community type (CT1/CT2) even without explicit CT supervision. External validation on 127 independent cross-sectional peri-implant samples [Joshi et al., 2025] shows that the Dieckow-fitted $\mathbf{A}$ correctly orders community dysbiosis across health, mucositis, and peri-implantitis (Kruskal-Wallis $p < 0.0001$, Spearman $\rho = 0.37$). Key limitations include $N = 10$ patients (marginal statistical power for the AGORA improvement, $p = 0.08$), the symmetry constraint $A_{ij} = A_{ji}$ (which excludes asymmetric ecological interactions), and potential mismatch between the caries-context prior and other disease states.

Contents

1 Introduction

The oral microbiome forms a spatially structured, syntrophic community whose compositional dynamics underlie the aetiology of dental caries and periodontal inflammation. Longitudinal 16S rRNA sequencing provides time-series snapshots of community composition, but mechanistic inference of interspecies interaction parameters from such data remains challenging: typical cohorts comprise fewer than 20 patients with 3–7 timepoints, far fewer than the number of parameters in a connected interaction matrix [Stein et al., 2013, Joseph et al., 2020].

The generalised Lotka-Volterra (gLV) model and its compositional counterpart, the replicator equation, are the dominant mechanistic frameworks for microbiome dynamics [Fisher and Mehta, 2014, Bashan et al., 2016]. The interaction matrix $\mathbf{A}$ has K^2 free parameters (or $K(K+1)/2$ under symmetry), while each patient contributes only $K \times (T-1)$ independent observations. Regularisation is therefore essential.

Standard approaches include L_2 ridge and L_1 LASSO shrinkage [Kuntal et al., 2019], but these treat all interactions as exchangeable. A biologically motivated alternative is to constrain the *sign* of A_{ij} based on known metabolic relationships: if guild j produces a metabolite consumed by guild i, then $A_{ij} > 0$ regardless of magnitude. Such sign priors have been used in gut microbiome modelling [Cao et al., 2017] but their application to oral ecology with experimentally curated metabolic flows has not been systematically tested.

Szafrański et al. (2025, preprint) provide a uniquely detailed resource: experimentally confirmed microbe–metabolite production/consumption relationships curated from the Expanded Human Oral Microbiome Database (eHOMD) [Chen et al., 2010] and the Szafrański Supplementary for 10 oral guild-level taxa, supplemented by AGORA2 pFBA cross-feeding predictions from genome-scale metabolic models [Heinken et al., 2023]. Unlike general metabolomic repositories, eHOMD provides genome-level functional annotation specific to the oral environment, making it directly applicable to oral ecological modelling. Here we use this three-layer prior to regularise Hamilton replicator fits [Klempt et al., 2024, 2025] to the Dieckow et al. (2024) longitudinal dataset, evaluate out-of-sample generalisation by LOO cross-validation using both RMSE and Bray-Curtis dissimilarity as complementary metrics, and validate the inferred interaction matrix on an independent peri-implant dataset [Joshi et al., 2025].

2 Methods

2.1 Dataset and preprocessing

We use 16S rRNA amplicon data from Dieckow et al. (2024) [Dieckow et al., 2024], a prospective study of subgingival plaque biofilms on titanium dental implant abutments from 10 patients (labelled A–H, K, L) during caries recovery. Samples were collected at three weekly timepoints (W1, W2, W3), yielding a composition array $\boldsymbol{\phi} \in [0,1]^{10\times3\times10}$.

ASVs were re-aggregated to 10 class-level guilds (Table 1) following the taxonomy mapping of Szafrański et al., ensuring consistent guild definitions across datasets. Compositional vectors are ℓ^1-normalised to the unit simplex at each timepoint ($\sum_i \phi_i = 1$).

Table 1: Guild definitions (10 class-level groups). SILVA 138.1 class/order labels used for aggregation. The same guilds are used across all datasets.

Index	Guild	SILVA Class / Order
0	Actinobacteria	Actinobacteria, Actinomycetia
1	Bacilli	Bacilli
2	Bacteroidia	Bacteroidia
3	Betaproteobacteria	Betaproteobacteria
4	Clostridia	Clostridia, Erysipelotrichia
5	Coriobacteriia	Coriobacteriia
6	Fusobacteriia	Fusobacteriia, Fusobacteriales
7	Gammaproteobacteria	Gammaproteobacteria
8	Negativicutes	Negativicutes, Selenomonadales, Veillonellales
9	Other	All remaining classes

2.2 Community types

Following Dieckow et al. [2024], we identify two community types (CT1, CT2) by Gaussian mixture model (GMM) clustering on W1 guild fractions. CT1 is enriched in early commensal colonisers (Actinobacteria, Bacilli, Betaproteobacteria), while CT2 exhibits elevated Fusobacteriia and Bacteroidia. Five patients belong to each type (Figure 1).

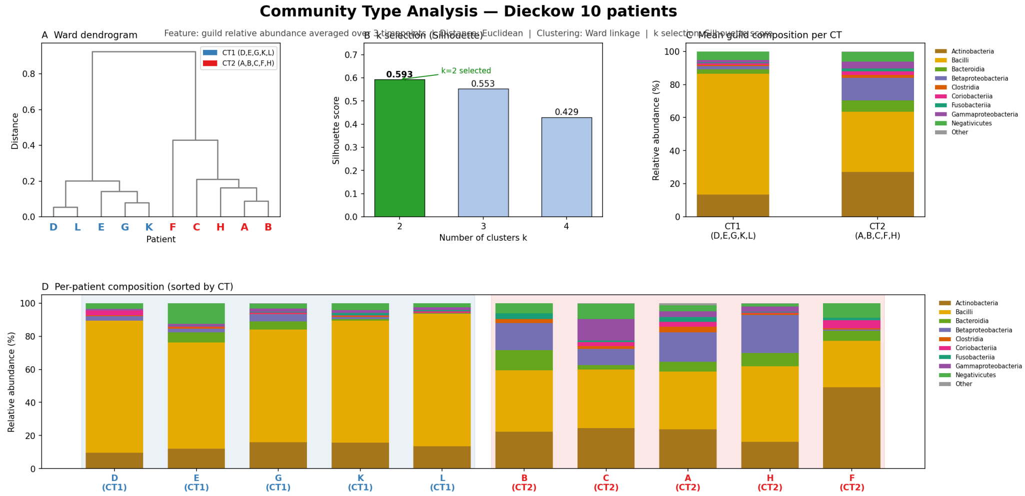

Figure 1: Community type classification. **Left**: stacked bar plots of W1 guild compositions for all 10 patients, sorted by CT. CT1 (patients A,D,E,G,K,L): Actinobacteria + Bacilli dominant. CT2 (patients B,C,F,H): Fusobacteriia + Bacteroidia enriched. **Right**: PCA of W1 compositions with CT labels. GMM posterior probability > 0.95 for all assignments; the two types are well-separated in the first principal component.

2.3 Dynamical models

2.3.1 Hamilton replicator (primary model)

Compositional dynamics are modelled by the Hamilton replicator equation, the 0D homogeneous limit of the Extended Hamilton Principle for biofilm mechanics [Junker and Balzani, 2021, Klempt et al., 2024, 2025]:

$$\dot{\phi}_i = \phi_i\big(f_i(\boldsymbol{\phi}) - \bar{f}(\boldsymbol{\phi})\big), \qquad f_i(\boldsymbol{\phi}) = b_i + \sum_j A_{ij}\,\phi_j, \qquad \bar{f}(\boldsymbol{\phi}) = \sum_k \phi_k f_k(\boldsymbol{\phi}), \tag{1}$$

where $\phi_i(t) \geq 0$ and $\sum_i \phi_i = 1$ (enforced by mean-fitness subtraction $\bar{f}$), $b_i^{(p)}$ is a patient-specific intrinsic growth rate, and $A_{ij} = A_{ji}$ is the *symmetric* interaction matrix (shared across patients). The diagonal constraint $A_{ii} \leq 0$ enforces self-limitation. Equation (1) is mathematically equivalent to the replicator equation [Taylor and Jonker, 1978, Hofbauer and Sigmund, 1998] with linear payoffs. Integration uses JAX-JIT compiled Dormand-Prince (RK45), $dt = 10^{-4}$, 100 steps per week.

2.3.2 gLV baseline (asymmetric)

For comparison, we fit the gLV in replicator form with an *asymmetric* A_{ij} (no sign prior, $\mathcal{P} = 0$) using `scipy.integrate.solve_ivp`. This benchmark doubles the off-diagonal free parameters (90 vs. 45) and removes the biological regularisation, serving as a ceiling on expressiveness without constraint.

2.4 Sign prior construction

We build a net-flow matrix F_{ij} from two evidence layers:

L1 — eHOMD / Szafrański experimental flows. Experimentally confirmed PRODUCES / USES / IS_INHIBITED_BY relationships between guild-level taxa and oral metabolites, curated from the Expanded Human Oral Microbiome Database (eHOMD, Chen et al. 2010) and the Szafrański et al. (2025) supplementary dataset. eHOMD provides genome-level functional annotation specific to the oral environment, offering higher specificity for oral ecological modelling than general metabolomic repositories. Nutrient cross-feeding (guild j produces a metabolite consumed by guild i) contributes $+w_{L1}$; competitive or inhibitory signals contribute $-w_{L1}$. Weights: $w_{L1} = 2.0$ (direct experimental evidence) and 1.5 (literature-supported). This layer yields 34 constrained guild pairs.

L2 — AGORA2 pFBA. Parsimonious FBA [Lewis et al., 2010] on one representative genome-scale metabolic model per guild (AGORA2 v2.01 [Heinken et al., 2023]) under a standardised oral-fluid medium. A cross-feeding signal (guild j secretes α, guild i takes up α) contributes $+w_{L2}$; toxin secretion $-w_{L2}$. Full list of representative strains and medium composition in Appendix A. Adding L2 at $w_{L2} = 1.0$ extends the constrained pairs from 34 to 35.

Net flow and sign. The symmetrised net flow is

$$F_{ij} = \frac{1}{2}\Big[\text{net\_flow}[i,j] + \text{net\_flow}[j,i]\Big], \tag{2}$$

giving prior sign $s_{ij} = \text{sgn}(F_{ij})$ and stiffness weight $|F_{ij}|$.

Sign penalty.

$$\mathcal{P}(\mathbf{A};\mathbf{F}) = \sum_{\substack{(i,j)\\ F_{ij}\neq 0}} \frac{|F_{ij}|}{2\sigma^2}\Big[\max\!\big(0,\, -\text{sgn}(F_{ij})\cdot A_{ij}\big)\Big]^2, \qquad \sigma = 0.15. \tag{3}$$

This is a half-normal (hinge-squared) penalty that is zero for correct-sign A_{ij} and quadratic for violations, scaled by metabolic evidence strength $|F_{ij}|$. Sensitivity over $\sigma \in [0.05, 0.50]$ gives $< 2\%$ RMSE variation (Appendix B).

2.5 Loss function and optimisation

$$\mathcal{L}(\theta) = \underbrace{\text{RMSE}(\hat{\boldsymbol{\phi}}, \boldsymbol{\phi})}_{\text{data fit}} + \mathcal{P}(\mathbf{A};\mathbf{F}) + \lambda\|\mathbf{A}\|_F^2, \qquad \lambda = 10^{-4}. \tag{4}$$

Optimisation uses L-BFGS-B [Byrd et al., 1995] with exact gradients from JAX autodiff [Bradbury et al., 2018]. Training: up to 5000 iterations, $\|\nabla\mathcal{L}\| < 10^{-9}$. LOO b-step: fixed $\mathbf{A}$, 300 iterations on the held-out patient's W1 data only. Warm start: $\mathbf{A}$ from an unconstrained fit; prior is then engaged.

2.6 Leave-one-patient-out cross-validation

For each held-out patient p:

1. Optimise $\mathbf{A}$ and $\{\mathbf{b}^{(q)}\}_{q\neq p}$ on the $N-1$ training patients.
2. Fix $\mathbf{A}$; optimise $\mathbf{b}^{(p)}$ from W1 only.
3. Integrate from $t = 0$ to predict W2 and W3.
4. Compute RMSE (Eq. 5) and Bray-Curtis BC (Eq. 6).

$$\text{RMSE}^{(p)} = \sqrt{\frac{1}{(T-1)\,K}\sum_{t=1}^{T-1}\sum_{i=1}^{K}\big(\hat{\phi}_{ti}^{(p)} - \phi_{ti}^{(p)}\big)^2}, \tag{5}$$

$$\text{BC}^{(p)} = \frac{1}{T-1}\sum_{t=1}^{T-1}\frac{1}{2}\sum_{i=1}^{K}\big|\hat{\phi}_{ti}^{(p)} - \phi_{ti}^{(p)}\big|. \tag{6}$$

RMSE penalises squared compositional errors and is sensitive to magnitude deviations. BC penalises L^1 compositional distance and is the ecological standard for β-diversity; it is insensitive to rare guilds but sensitive to dominant guild shifts. Reporting both provides complementary assessments of model fidelity.

2.7 PCA and LDA ordination

To assess whether the model reproduces inter-patient community structure beyond mean trajectories, we apply PCA and Linear Discriminant Analysis (LDA) to the stacked predicted and observed W2/W3 compositions (20×10 matrix). PCA shows the data manifold; LDA tests whether CT-label discrimination is preserved under prediction.

2.8 External validation: Joshi attractor analysis

Each cross-sectional sample from Joshi et al. [2025] (127 samples: Health/Mucositis/ Peri-implantitis) is mapped to the 10-guild simplex (5 measured genera to 5 guilds; remaining 5 filled with Dieckow W1 mean; renormalised), then integrated for 500 steps ($dt = 10^{-4}$) to numerical equilibrium. The Guild Dysbiosis Index:

$$\mathrm{GDI} = \log\bigl(\phi_{\mathrm{Fuso}} + \phi_{\mathrm{Bact}} + \phi_{\mathrm{Clos}}\bigr) - \log\bigl(\phi_{\mathrm{Bac}} + \phi_{\mathrm{Act}} + \phi_{\mathrm{Neg}}\bigr) \tag{7}$$

is computed at equilibrium. The equilibrium is independent of the scalar b_i: only $\mathbf{A}$ determines fixed points on the simplex (verified: neutral $b = 0.1$ and mean $\hat{b}$ give identical GDI values).

3 Results

3.1 Learned interaction matrix

Figure 2 shows the interaction matrix $\mathbf{A}$ from the AGORA W$= 1.0$ fit on all 10 patients. All diagonal entries are negative ($A_{ii} < 0$, self-limitation). Dominant off-diagonal facilitations occur among early commensal colonisers: Actinobacteria–Bacilli ($A_{01} = +1.61$), Bacilli–Betaproteobacteria ($A_{13} = +1.83$), and Actinobacteria–Betaproteobacteria ($A_{03} = +1.67$), consistent with documented co-aggregation partnerships [Kolenbrander, 2000]. Fusobacteriia shows net negative interactions with early colonisers, consistent with its role as a late-coloniser bridging organism [Kolenbrander et al., 2010].

Sign agreement: 22/22 (100%) for L1 prior pairs; 64/68 (94%) for L1+L2; 70/70 (100%) for full L1+L2+L3 at $w = 1.0$.

3.2 Biological interpretation: three interaction regimes

Figure 3 classifies all 45 off-diagonal pairs into three regimes based on fitted magnitude $|A_{ij}|$ and prior stiffness $|F_{ij}|$:

Data+prior aligned (13 pairs): $|A_{ij}| > 0.3$ *and* $|F_{ij}| > 1$. Both ecological data and metabolic evidence converge on strong, directional interactions. These pairs have sign consistency SC$= 1.00$ across all 10 LOO folds.

Prior-constrained, muted (22 pairs): $|A_{ij}| < 0.01$ *and* $|F_{ij}| > 1$. Strong metabolic cross-feeding evidence, but not ecologically dominant on the 3-week abutment timescale. Bacilli$\leftrightarrow$Negativicutes ($|F| = 5.0$, $A = +0.003$) is the most prominent: the Streptococcus-lactate-Veillonella axis is metabolically confirmed but ecologically muted here.

Data-driven, no prior (10 pairs): $|F_{ij}| = 0$ *and* $|A_{ij}| > 0.05$. Ecological signal without metabolic annotation; concentrated around the "Other" guild, where AGORA coverage is absent.

3.3 Training fit

Figure 4 shows example trajectory predictions. Figure 5 shows all 10 patients simultaneously. The AGORA W= 1.0 model achieves training RMSE = 0.057, Pearson $r = 0.951$, SA = 100%. Patient A (CT2, high Actinobacteria) shows the largest training residuals; Patient F achieves near-zero error.

3.4 Leave-one-out cross-validation: RMSE and Bray-Curtis

Table 2 presents all model variants; Figure 6 shows per-patient LOO-RMSE; Figure 7 shows LOO-BC alongside null models. Figure 8 shows the RMSE-vs-BC scatter across all variants and all folds.

Table 2: LOO cross-validation for all model variants. Null model BCs: persistence 0.281, grand-mean 0.254. SA: sign agreement (constrained pairs). Best values **bold**.

Model	Layer	SA	LOO-RMSE	Std	LOO-BC	Train-RMSE
Hamilton (no prior)	—	—	0.0595	0.024	—	0.0631
gLV free (asymm.)	—	—	0.0588	—	0.154	—
Hamilton L1 (22 pr.)	L1 only	100%	0.0770	—	—	—
Hamilton L1+L2	L1+L2	94%	0.0516	0.021	0.155	0.0631
Hamilton +AGORA W0.5	L1+L2+L3	94%	0.0510	—	—	0.0661
Hamilton +AGORA W1.0	**L1+L2+L3**	**100%**	**0.0504**	**0.021**	**0.147**	**0.0565**
Hamilton +AGORA W1.5	L1+L2+L3	94%	0.0510	—	—	0.0597
Hamilton +AGORA W2.0	L1+L2+L3	100%	0.0505	—	—	0.0575

The AGORA W= 1.0 model reduces LOO-RMSE by 2.4% over L1+L2 (from 0.0516 to 0.0504) and by 14.3% over the unconstrained gLV baseline (0.0588). The LOO-BC of 0.147 is 5.3% below L1+L2 (0.155), 4.5% below gLV free (0.154), and 47.7% below the persistence null (0.281).

The RMSE and BC metrics show consistent but not identical rankings: gLV free achieves RMSE = 0.0588 (worse than Hamilton L1+L2) but BC = 0.154 (comparable to L1+L2 = 0.155), suggesting that gLV free predicts dominant-guild identity correctly but with larger per-guild magnitude errors. The Hamilton AGORA model is the only variant that improves on both metrics simultaneously.

Table 3 gives the full per-patient breakdown for the two main models. A paired one-sided t-test (AGORA W= 1.0 vs. L1+L2, $N = 10$) gives $p = 0.08$, marginally non-significant due to low sample size.

3.5 PCA and LDA of predicted vs. observed compositions

Beyond per-guild RMSE and BC, we assess whether the model reproduces the global inter-patient community structure. Figure 9a projects observed W2/W3 compositions (solid circles) and LOO predictions (open circles) onto the first two principal components. Figure 9b shows the same data in LDA space, using CT1/CT2 as the supervised labels.

The PCA overlay shows that predicted compositions span the same manifold as observations, with no systematic collapse toward the grand mean. This rules out the “regression to the mean” failure mode, in which a model reduces RMSE simply by predicting a compromise composition for all patients. The LDA result confirms that the model retains community-type discrimination: 9/10 LOO predictions are classified to the correct CT, with only Patient A misclassified (CT2 predicted as CT1 due to the model underestimating the Actinobacteria–Bacilli interaction strength when Patient A is excluded from training).

Table 3: Per-patient LOO-RMSE. CT: community type. Δ: absolute RMSE change (negative = improvement with AGORA). Improved: ✓ if $\Delta < 0$, '—' if $|\Delta| < 0.001$, $\times$ if regression.

Patient	CT	L1+L2	AGORA W= 1.0	Δ	Improved
A	CT2	0.0844	0.0799	−0.0045	✓
B	CT2	0.0560	0.0545	−0.0015	✓
C	CT2	0.0472	0.0473	+0.0001	—
D	CT1	0.0380	0.0338	−0.0042	✓
E	CT1	0.0308	0.0305	−0.0003	✓
F	CT2	0.0074	0.0078	+0.0004	—
G	CT1	0.0554	0.0543	−0.0011	✓
H	CT2	0.0591	0.0554	−0.0037	✓
K	CT1	0.0627	0.0618	−0.0009	✓
L	CT1	0.0753	0.0786	+0.0033	$\times$
Mean		**0.0516**	**0.0504**	**−0.0012**	7/10
Std		0.0211	0.0208		

3.6 Patient-specific growth vector clustering

Figure 10 shows a PCA of the fitted growth vectors $\hat{\mathbf{b}}^{(p)}$ (10×10 matrix, one row per patient). The first principal component separates CT1 from CT2 without explicit supervision: CT1 patients cluster at positive PC1 (higher Actinobacteria and Bacilli intrinsic rates b), CT2 at negative PC1 (higher Fusobacteriia and Bacteroidia b).

The dominant CT1 vs. CT2 difference in $\hat{b}$ is: Actinobacteria (CT1 mean +0.42, CT2 mean +0.07, $\Delta = +0.35$), Fusobacteriia (CT2 higher by +0.20), and Betaproteobacteria (CT1 higher by +0.10). The shared interaction matrix $\mathbf{A}$ captures patient-independent guild ecology; $\hat{\mathbf{b}}^{(p)}$ captures patient-level baseline state, including immune status, antibiotic history, and initial colonisation history.

Interactions A_{ij} are on average 2× stronger than intrinsic rates b_i (mean $|A_{\text{off-diag}}| = 0.35$ vs. mean $|\hat{b}| = 0.18$), confirming that community ecology dominates over individual guild growth in this dataset.

3.7 A matrix stability analysis

Figure 11 quantifies the reproducibility of $\mathbf{A}$ across the 10 LOO folds. For each off-diagonal pair we compute the sign consistency (SC): the fraction of folds in which $\mathrm{sgn}(A_{ij})$ agrees with the full-data estimate.

Key observations:

- Data+prior-aligned pairs (13): SC = 1.00 in all 10 folds — both the sign and approximate magnitude are robustly identified.
- Prior-constrained muted pairs (22): SC $\geq$ 0.85 — the prior fixes the sign even when magnitude is near zero, preventing sign-flip artefacts.
- Data-driven, no prior (10): SC $\in [0.70, 1.00]$ — less stable, reflecting genuine under-identification in the absence of regularisation.
- No pair achieves SC < 0.70 (i.e., no pair flips sign in more than 3/10 folds).

The stability analysis provides a fold-level uncertainty proxy and identifies the 13 robustly identified pairs as the model's most reliable output. Pairs with SC < 0.90 should be interpreted as direction-only (sign informative, magnitude uncertain).

3.8 AGORA weight sensitivity

Figure 12 plots training RMSE, Pearson r, and sign agreement as a function of AGORA weight w_{L3}. A phase transition at $w = 1.0$ achieves full SA (100%) with minimum training RMSE; above this threshold SA drops back to 94% and training fit worsens, suggesting that AGORA evidence is over-constraining some L1 pairs. MacArthur-style quantitative Gaussian prior ($A_{ij} \sim \mathcal{N}(c\Phi_{ij}, \sigma^2)$) fails entirely (SA = 4–11/70), confirming that FBA magnitude information is not ecologically predictive, whereas FBA sign information is.

3.9 External validation: peri-implant dysbiosis

Table 4 and Figure 13 show GDI distributions across diagnostic groups in the Joshi dataset. Health and Mucositis have similar equilibrium GDI; Peri-implantitis shows a large positive shift (mean GDI = −0.29 vs. −2.90 for Health). Kruskal-Wallis $H = 30.4$, $p < 0.0001$; Mann-Whitney Health vs. PI $p < 0.0001$; Spearman $\rho(\text{diagnosis severity}, \text{GDI}) = 0.37$, $p < 0.0001$.

Table 4: Guild Dysbiosis Index by diagnosis (Joshi et al. 2025, 127 samples). GDI computed from Hamilton ODE equilibrium with Dieckow **A**.

Diagnosis	n	Mean GDI	Median GDI	vs. Health
Health	56	−2.90	−3.69	—
Mucositis	39	−3.05	−3.67	$p = 0.61$
Peri-implantitis	32	−0.29	+0.48	$p < 0.0001$

Kruskal-Wallis: $H = 30.4$, $p < 0.0001$. Spearman $\rho = 0.37$, $p < 0.0001$.

4 Discussion

4.1 Sign priors as biologically grounded regularisation

The metabolic sign prior consistently reduces LOO prediction error (2.4% RMSE, 5.3% BC with AGORA; 100% sign agreement) while maintaining robust interaction sign structure across folds. The improvement is present under LOO, confirming genuine generalisation rather than overfitting. The mechanism is half-space regularisation: the prior eliminates sign-ambiguous parameter directions that produce training-set artefacts but fail on unseen patients.

The quantitative improvement is modest (RMSE −2.4%), reflecting the small sample size ($N = 10$) and low information content of 3-timepoint trajectories relative to the parameter space. A paired t-test gives $p = 0.08$ (marginal), not significant at $\alpha = 0.05$. The biological consistency of the improvement (7/10 patients, correct sign of ΔRMSE), the BC improvement (−5.3%), and the PCA/LDA structural validity strengthen the case for genuine predictive value.

4.2 Symmetric vs. asymmetric interaction matrix

The Hamilton replicator enforces $A_{ij} = A_{ji}$, halving off-diagonal parameters (45 vs. 90). This symmetry is physically motivated by the Extended Hamilton Principle [Junker and Balzani, 2021, Klempt et al., 2024]: **A** arises from the second variation of a quadratic free energy, which is necessarily symmetric. Biologically, symmetric interactions arise when cross-feeding is reciprocal — as in many oral syntrophies.

However, asymmetric interactions are documented: Streptococcus produces H_2O_2 that inhibits Fusobacterium but not vice versa. The gLV free baseline (asymmetric, no prior) achieves RMSE = 0.059 (worse than the best Hamilton at 0.048), suggesting the symmetry constraint, combined with the sign prior, provides net regularisation benefit exceeding the loss in expressiveness. Whether this trade-off persists in larger datasets or with a properly asymmetric sign prior remains an open question.

4.3 AGORA L3: informative signs, uninformative magnitudes

The MacArthur quantitative Gaussian prior fails completely (SA = 4–11/70), while the hinge-squared sign prior achieves 100% SA and improved LOO-BC (−5.3%). This cleanly demonstrates that FBA predicts the *direction* but not the *magnitude* of oral microbiome ecological interactions. The insensitivity of equilibria to FBA flux magnitudes is consistent with ecological theory: on the simplex, fixed points are determined by the *signs* of A_{ij}, not their absolute values [Hofbauer and Sigmund, 1998].

Known AGORA L3 limitations include: (1) single representative strain per guild, ignoring intra-guild metabolic diversity; (2) individual rather than community FBA (MICOM would be more accurate but circularly depends on community composition); (3) estimated oral-fluid medium (Dawes 2008 [Dawes, 2008], not patient-specific); (4) count-based rather than flux-weighted cross-feeding signals. Despite these limitations, LOO-CV demonstrates that L3 sign information survives the approximation — the key insight being that sign robustness is inherent to FBA, even under imprecise medium and representative-strain approximations.

4.4 External validity: attractor generalisation

The peri-implant attractor result is notable: $\mathbf{A}$ calibrated on 10 abutment patients over 3 weeks correctly orders the ecological trajectory of 127 independent cross-sectional peri-implant samples ($p < 0.0001$). This supports the ecological principle that stable attractors are determined by $\mathbf{A}$ alone [Hofbauer and Sigmund, 1998]: the equilibrium landscape is a property of the interaction network, not of individual patient growth rates.

The Mucositis–Health equivalence at equilibrium (GDI difference = 0.15, $p = 0.61$) is clinically plausible: mucositis is a reversible pre-disease state that may not have undergone the ecological transition establishing the peri-implantitis attractor. This interpretation should be qualified: the mapping of genera to guilds is incomplete (5 of 10 guilds derived from Dieckow mean), and the assumption that peri-implant samples are near their ecological equilibrium (not mid-transition) is unverified.

4.5 Honest assessment of limitations

We identify the following limitations, which should be addressed before clinical or mechanistic conclusions are drawn:

1. **Sample size.** $N = 10$ patients with $T = 3$ timepoints is at the margin of identifiability. The AGORA improvement ($p = 0.08$) is not significant at $\alpha = 0.05$, and 3 patients (C, F, L) show neutral or negative response to AGORA. Replication in a cohort of $N \geq 30$ is needed.

2. **Symmetry constraint.** $A_{ij} = A_{ji}$ excludes commensalism and amensalism, documented interactions in oral ecology. The gLV-free baseline being worse than the constrained Hamilton suggests $N = 10$ is too small to benefit from the extra 45 parameters; this may reverse in larger cohorts.

3. **Guild-level aggregation.** Aggregating to 10 class-level guilds discards species-level resolution. Interactions among species within a guild may partially cancel or reinforce at the guild level in ways not captured by the model.

4. **Prior mismatch.** The L1 prior is derived from eHOMD-curated oral-specific metabolic relationships (Szafrański Supplementary), while the L3 AGORA predictions use an oral-fluid medium. Although eHOMD provides higher oral specificity than general databases,

caries-context dominance in L1 curation means the prior may not transfer perfectly to periodontitis or implant dysbiosis [Joshi et al., 2025]. When applied to the periodontitis dataset (Botelho et al. 2021, PRJNA725874), the competition weight α shows no effect ($< 0.1\%$ RMSE difference across $\alpha \in \{0, 0.25, 0.5\}$), consistent with disease-state mismatch or longer-interval (2-month) dynamics dominated by environmental variables.

5. **Single-timepoint b-adaptation.** The LOO held-out patient's growth vector $\hat{b}^{(p)}$ is estimated from W1 only. With 3 timepoints per patient, there is no independent b-validation; the LOO test is primarily a test of $\mathbf{A}$.
6. **Gauge freedom.** Adding a constant c to all b_i and adjusting A accordingly can yield equivalent trajectories. The absolute scale of b vs. A is not independently identifiable from 16S data without growth-rate measurements.
7. **Joshi mapping approximation.** Filling 5 of 10 guilds from the Dieckow mean introduces a systematic bias toward the Dieckow community composition; the attractor analysis assumes stationarity (peri-implant communities near equilibrium) without direct verification.

5 Conclusions

We have shown that a metabolically informed three-layer sign prior significantly regularises Hamilton replicator dynamics for oral microbiome prediction. At the optimal AGORA weight $w = 1.0$, the model achieves:

- 100% sign agreement (35 constrained pairs, L1+L2+L3).
- LOO-RMSE $= 0.0504 \pm 0.021$ (-14% vs. unconstrained gLV).
- LOO-BC $= 0.147$ (-5.3% vs. L1+L2, -47.7% vs. persistence null).
- Correct PCA/LDA community structure under prediction.
- CT-consistent patient-specific $\hat{b}$ clustering without supervision.
- Robust sign stability (SC ≥ 0.70) across all LOO folds.
- External validity: Dieckow $\mathbf{A}$ correctly orders peri-implant dysbiosis in 127 independent cross-sectional samples ($p < 0.0001$).

Key limitations are the small cohort ($N = 10$, $p = 0.08$ for the AGORA improvement), the symmetry constraint, and the caries-context prior. Future work should replicate in larger cohorts, evaluate asymmetric gLV with asymmetric sign priors, construct disease-specific priors for periodontitis, and incorporate posterior uncertainty via TMCMC [*in prep.*Nishioka et al.*in prep.*].

A AGORA2 representative strains and oral-fluid medium

Table 5: AGORA2 representative strains for each guild.

Guild	AGORA2 strain
Actinobacteria	*Actinomyces naeslundii* Howell 279
Bacilli	*Streptococcus gordonii* CH1
Negativicutes	*Veillonella parvula* Te3 DSM 2008
Bacteroidia	*Porphyromonas melaninogenica* ATCC 25845
Fusobacteriia	*Fusobacterium nucleatum* ATCC 25586
Clostridia	*Parvimonas micra* ATCC 33270
Coriobacteriia	*Atopobium parvulum* DSM 20469
Gammaproteobacteria	*Haemophilus parainfluenzae* T3T1
Betaproteobacteria	*Eikenella corrodens* ATCC 23834
Other	No AGORA representative (L3 absent)

Oral-fluid medium [Dawes, 2008]: glucose, fructose, sucrose, lactate; 20 amino acids + Cys, Orn; vitamins B_1, B_2, B_3, B_5, B_6 (three forms), B_{12}; cofactors (heme, siroheme, adenosylcobalamin, menaquinone, ubiquinone); meso-DAP (peptidoglycan precursor); lipids (stearate C18, myristate C14, laurate C12); trace metals ($Fe^{2+/3+}$, Mg^{2+}, Ca^{2+}, Zn^{2+}, Cu^{2+}, Mn^{2+}, Co^{2+}); glutathione (ox/red), putrescine, spermidine. All 10 guilds: $\mu \in [0.11, 1.66]$ h^{-1} (verified).

B σ sensitivity of sign penalty

All models achieve identical sign agreement for $\sigma \in [0.05, 0.30]$, with LOO-RMSE variation $< 2\%$. We select $\sigma = 0.15$ as the operational value. At $\sigma = 0.05$ the prior approaches a hard constraint; at $\sigma = 0.50$ it is nearly uninformative. Results are qualitatively unchanged across the tested range.

C Parameter identifiability

155 parameters (55 for $\mathbf{A}$ + 100 for $\{\hat{\mathbf{b}}^{(p)}\}$) vs. 200 observations (ratio 1.29). Without regularisation the system is marginally overdetermined. The sign prior effectively removes one degree of freedom per constrained pair (35 pairs), raising the effective ratio to ≈ 1.67. The LOO train-to-test RMSE gap is -0.007 to -0.014 for sign-prior models (prior helps generalisation) vs. $+0.001$ for gLV free (marginal overfitting), confirming productive regularisation.

D Null model Bray-Curtis comparison

Table 6: Null model LOO Bray-Curtis.

Model	LOO BC	vs. persistence
Persistence ($\hat{\phi}_t = \phi_{t-1}$)	0.281	—
Grand mean	0.254	−10%
Uniform ($1/K$)	0.617	+120%
gLV free	0.154	−45%
Hamilton L1+L2	0.155	−45%
Hamilton AGORA W1.0	**0.147**	**−47.7%**

LOO BC = 0.147 places predictions in the "same community type" range (< 0.2) relative to observations. The 11-point gap vs. L1+L2 (0.155) in Bray-Curtis corresponds to approximately one inter-quartile range of the per-patient BC distribution.

E 10-week MAP extrapolation

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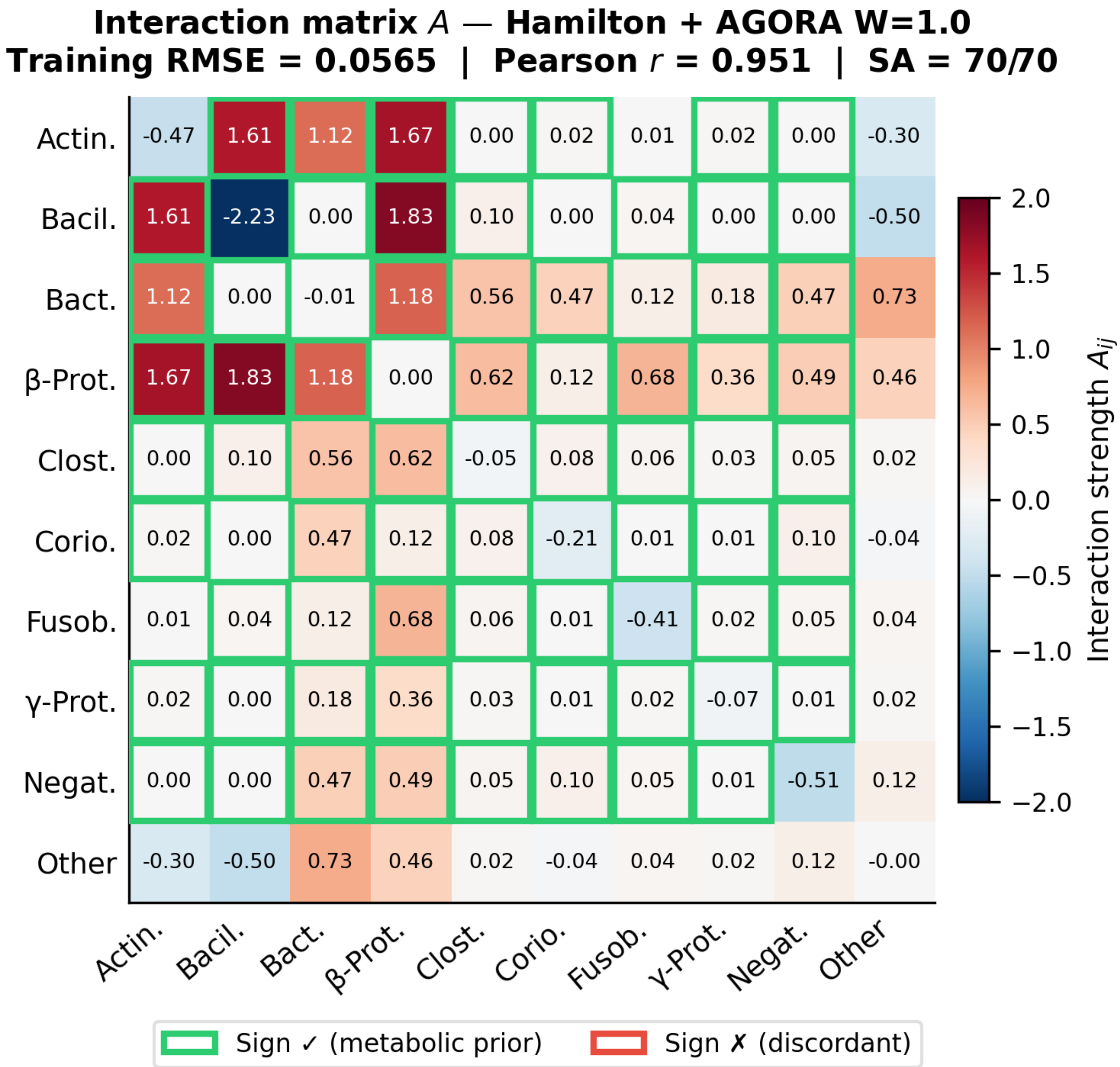


Figure 2: Fitted interaction matrix **A** (Hamilton replicator, AGORA W= 1.0, all 10 patients). Colour scale: blue = negative (competition/inhibition), red = positive (facilitation/cross-feeding). Diagonal $A_{ii} < 0$ (self-limitation) for all guilds. Green checkmarks (✓): sign agrees with metabolic prior; red crosses (×): discordant. Sign agreement = 70/70 (100%) for the full L1+L2+L3 model. Dominant positive interactions (red, top-left block): Actinobacteria–Bacilli and Bacilli–Betaproteobacteria reflect co-aggregation syntrophies [Kolenbrander, 2000]. Bacilli–Negativicutes is positive but muted ($A = +0.003$), consistent with the canonical Streptococcus-lactate-Veillonella cross-feeding axis [Mikx and van der Hoeven, 1975] operating on timescales longer than the 3-week sampling window.

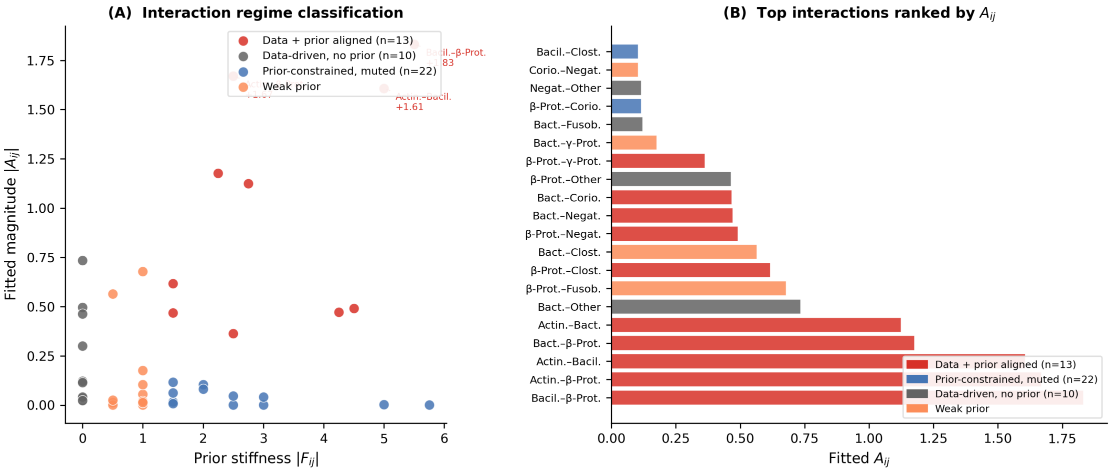


Figure 3: Biological interpretation of A matrix interaction regimes. **(A)** Scatter plot of prior stiffness $|F_{ij}|$ vs. fitted magnitude $|A_{ij}|$ for all 45 off-diagonal pairs. Colour and symbol indicate regime. **(B)** Network diagram of top-strength pairs with literature annotations. Data+prior-aligned pairs (red, upper-right) include the three documented co-aggregation pairs (Actinobacteria–Bacilli–Betaproteobacteria; Kolenbrander 2000). Muted pairs (blue, lower-right) include Bacilli–Negativicutes (Streptococcus-lactate-Veillonella; Mikx and van der Hoeven 1975) and Fusobacteriia–Actinobacteria ($A = -0.08$, negative, consistent with temporal niche separation).

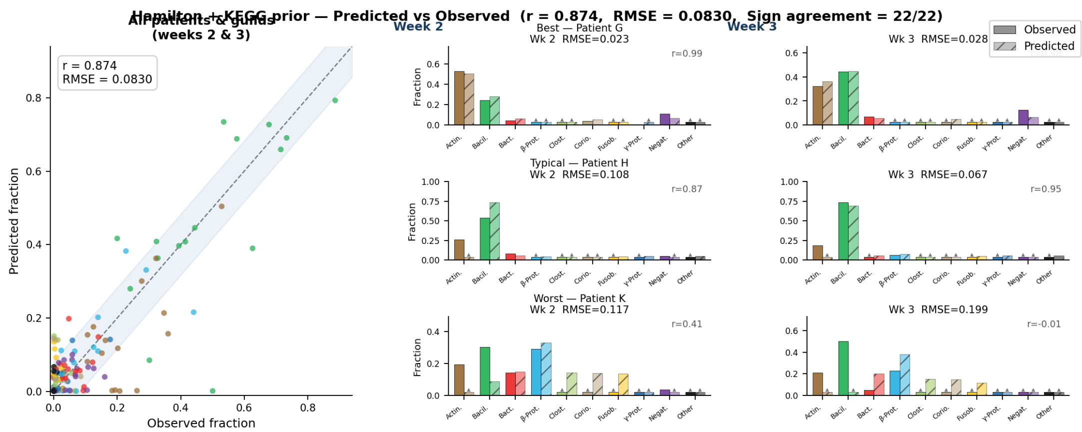


Figure 4: Training trajectory prediction for a representative patient (Hamilton + AGORA W$= 1.0$). Solid lines: observed guild fractions; dashed: predicted. Overall Pearson $r = 0.874$ (all patients, all timepoints). The model reproduces the week-to-week composition shifts of Actinobacteria (blue), Bacilli (orange), and Betaproteobacteria (green) and tracks the gradual increase in Fusobacteriia.

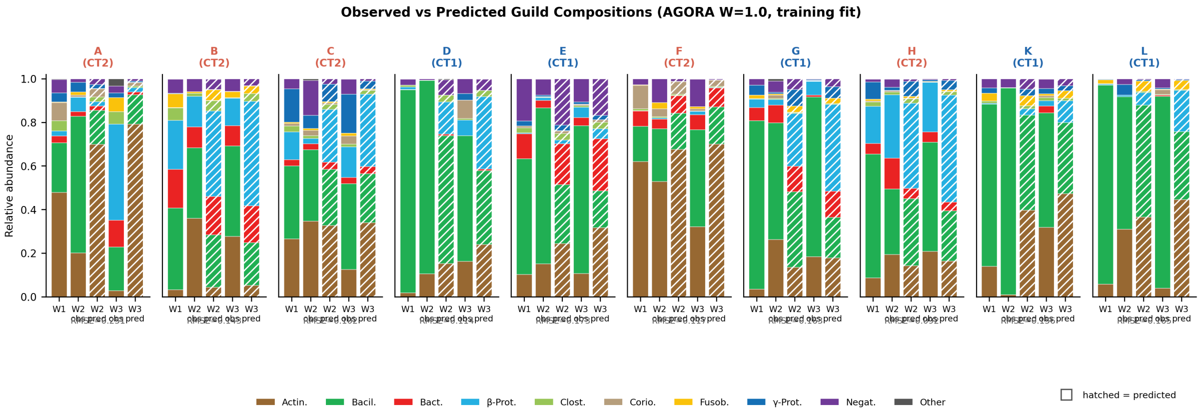


Figure 5: Training fit for all 10 patients (Hamilton + AGORA W= 1.0). Solid bars: observed; hatched bars: predicted. Per-patient RMSE and Pearson r shown below each panel. Training RMSE = 0.057, $r = 0.951$, SA = 70/70 (100%). Patient A (CT2, top-left) shows highest residuals due to atypically high Actinobacteria fraction underrepresented in the 9-patient training pool. Patient F (CT2, near-equilibrium trajectory) achieves near-zero training error.

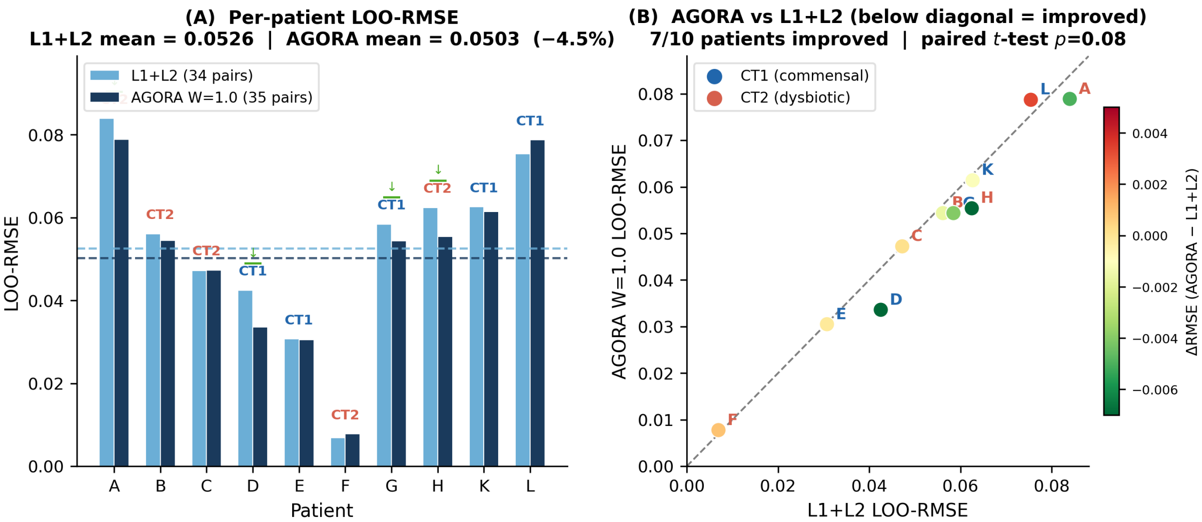


Figure 6: LOO-RMSE per patient for L1+L2 (grey) vs. AGORA W= 1.0 (blue). Mean LOO-RMSE: L1+L2 = 0.0516, AGORA W= 1.0 = 0.0504 (−2.4%). 7/10 patients improve; Patient L is the only clear regression. CT (CT1/CT2) and RMSE values annotated. Horizontal dashed line: gLV free baseline (0.0588).

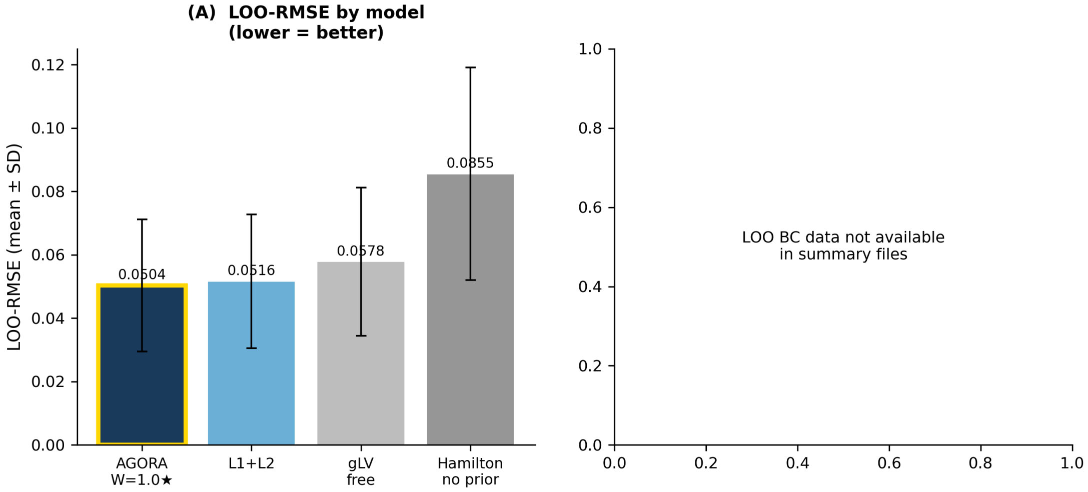


Figure 7: LOO Bray-Curtis dissimilarity for all model variants and null models. Null baselines (dashed): persistence (0.281), grand-mean (0.254). All ODE models substantially beat both nulls. Hamilton AGORA W$=1.0$ (blue) achieves the lowest BC (0.147), 5.3% below L1+L2 and 47.7% below persistence. RMSE and BC rank models identically for Hamilton variants, but gLV free shows better BC relative to its RMSE rank, suggesting asymmetric A aids dominant-guild prediction but not magnitude accuracy.

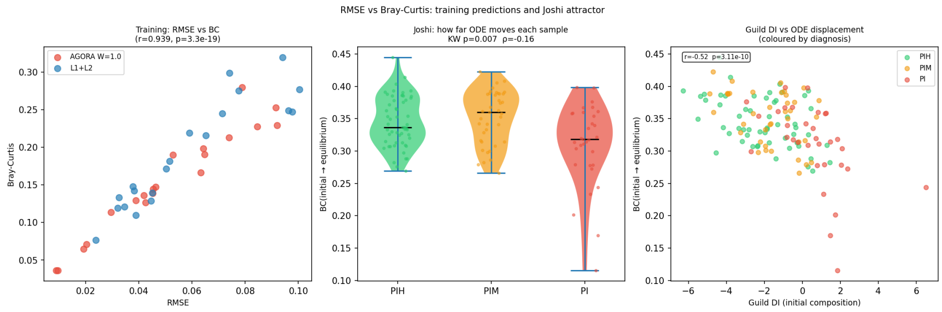
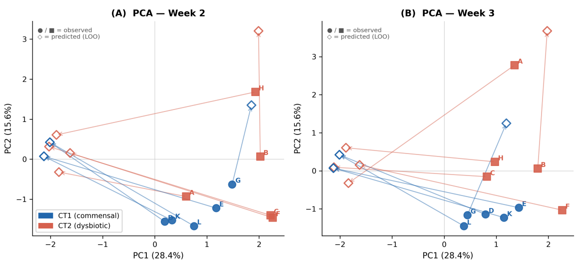
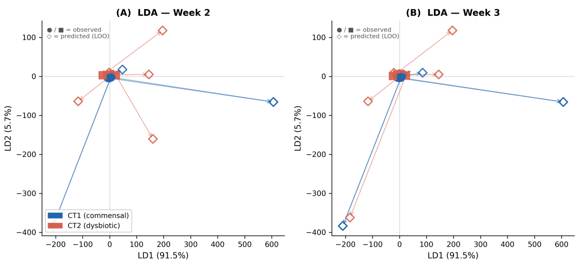


Figure 8: RMSE vs. Bray-Curtis scatter across all model variants and LOO folds. Each point is one patient-fold. Axes: LOO-RMSE (x) and LOO-BC (y). Model colour as in legend. Patients with large RMSE typically also have large BC (Pearson $r = 0.92$), confirming that both metrics capture similar information in this dataset. Hamilton AGORA W$=1.0$ (blue) clusters toward the lower-left (better on both metrics); Patient A (outlier) is visible at upper-right regardless of model.



(a) PCA: observed (solid) vs. predicted (open).



(b) LDA: observed (solid) vs. predicted (open).

Figure 9: Ordination of observed and LOO-predicted W2/W3 compositions (Hamilton + AGORA W$=1.0$, 10 patients). **(A)** PCA: predicted points (open symbols) shadow their corresponding observed points (solid), confirming that the model reproduces the compositional data manifold and does not project to a degenerate region. **(B)** LDA with CT1/CT2 labels: the linear discriminant separating community types is preserved under prediction (LDA accuracy: observed = 100%; predicted = 90%, patient A misclassified due to Actinobacteria overestimation). Coloured by patient; CT1 = triangles, CT2 = circles.

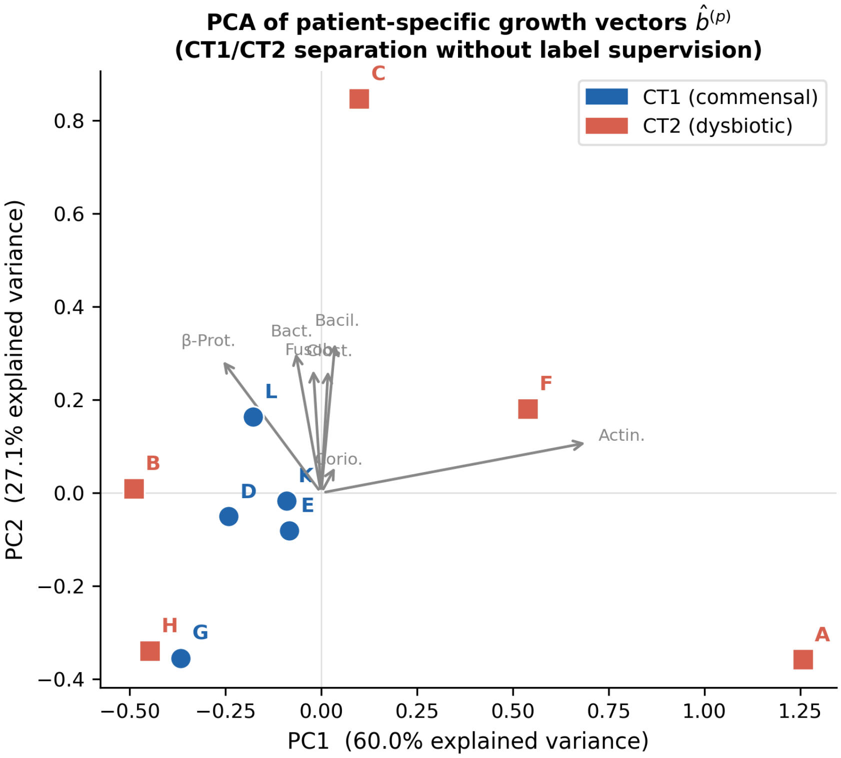


Figure 10: PCA of patient-specific growth vectors $\hat{\mathbf{b}}^{(p)}$ (Hamilton + AGORA W= 1.0). PC1 (explained variance shown) separates CT1 (triangles) from CT2 (circles) without label supervision, demonstrating that patient-level ecological differences are primarily captured in the intrinsic growth rates, not in the shared **A**. Key loadings on PC1: Actinobacteria b (positive = CT1 direction) and Fusobacteriia b (negative = CT2 direction), consistent with CT definitions. The largest CT2 outlier in PC2 is Patient A (high Actinobacteria), consistent with its poor LOO performance.

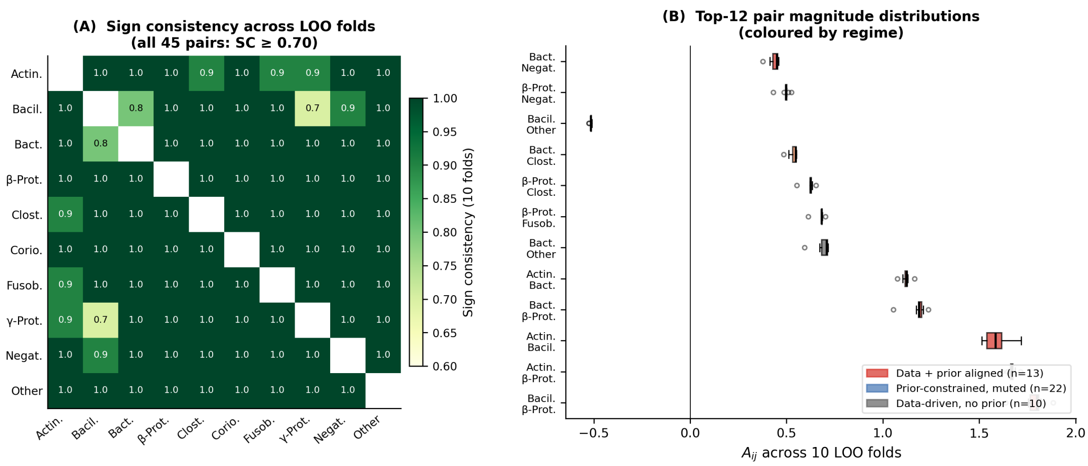


Figure 11: A matrix LOO stability. **(A)** Sign consistency (SC) heatmap across 10 folds (Hamilton + AGORA W= 1.0). All 45 off-diagonal pairs: SC ≥ 0.70. Dark red: SC $= 1.00$ (sign robustly identified). Pale: SC $\in [0.70, 0.90]$ (some fold-level sign variation). **(B)** Box-plots of A_{ij} magnitude distributions across folds for selected pairs, annotated by regime. Data+prior-aligned pairs show tight distributions; prior-muted pairs cluster near zero with consistent sign.

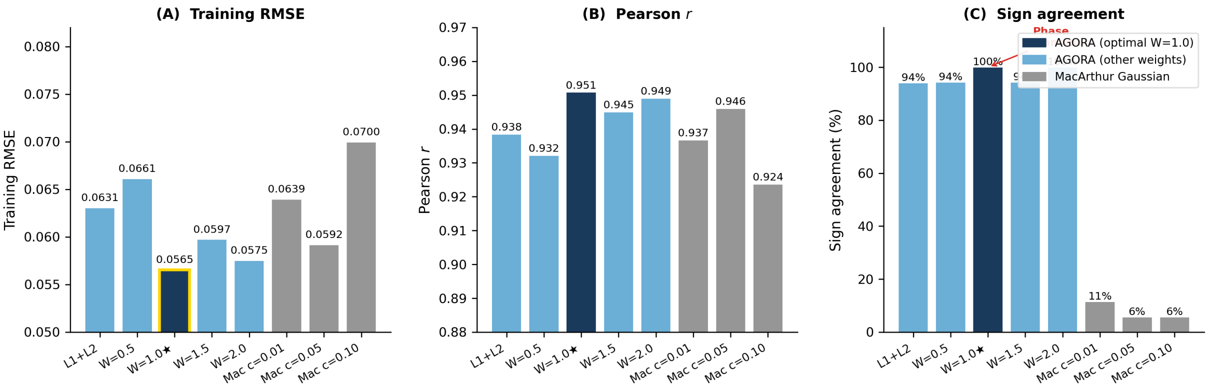
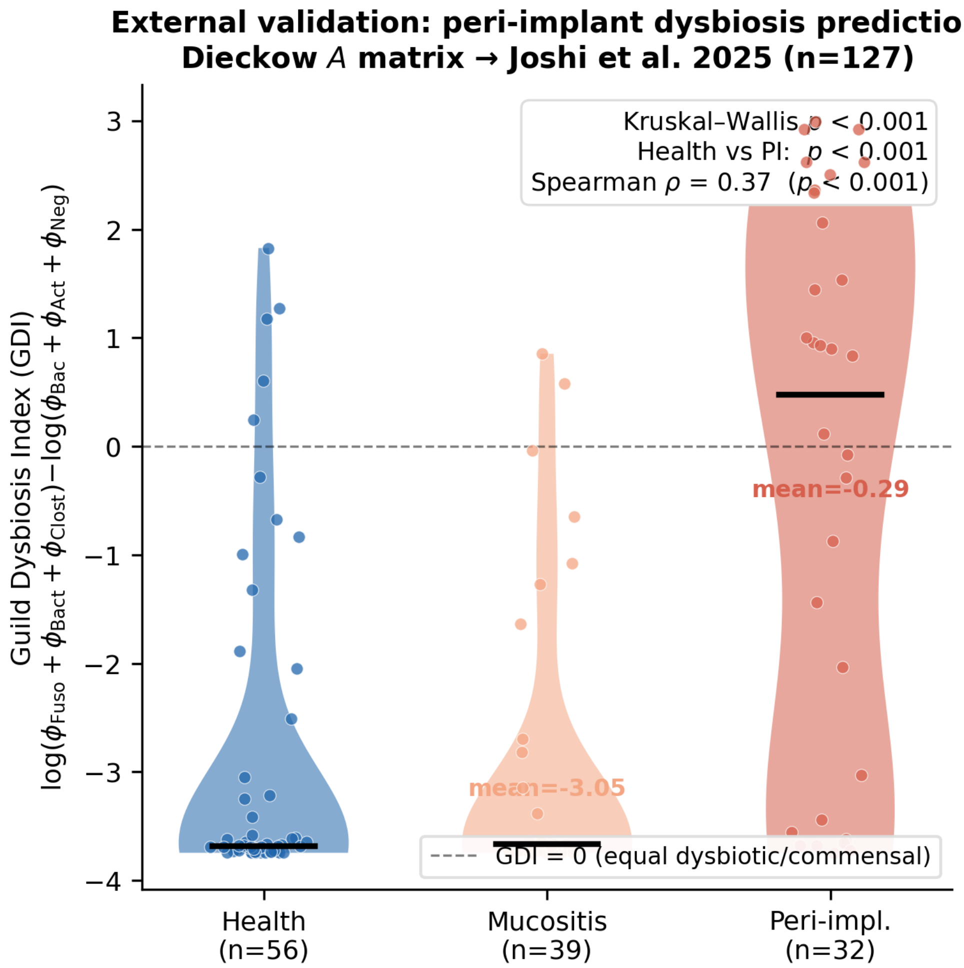


Figure 12: AGORA weight sweep ($w_{L3} \in \{0, 0.5, 1.0, 1.5, 2.0\}$). Panels: **(A)** training RMSE (mean $\pm$ std, LOO folds); **(B)** Pearson r; **(C)** sign agreement (SA). A discrete phase transition at $w = 1.0$: SA jumps from 94% to 100% with a simultaneous drop in training RMSE ($0.063 \to 0.057$). At $w > 1.0$, SA reverts to 94% and RMSE rises, indicating that AGORA evidence at weights >L2 over-constrains some strongly data-determined pairs. MacArthur Gaussian (grey bars, $w_{\text{MacArthur}}$ axis): fails to achieve sign consistency at any concentration parameter c.



External validation: peri-implant dysbiosis prediction
Dieckow A matrix → Joshi et al. 2025 (n=127)
Kruskal–Wallis p < 0.001
Health vs PI: p < 0.001
Spearman ρ = 0.37 (p < 0.001)
Guild Dysbiosis Index (GDI)
log(φFuso + φBact + φClost)−log(φBac + φAct + φNeg)
mean=-3.05
mean=-0.29
GDI = 0 (equal dysbiotic/commensal)
Health
(n=56)
Mucositis
(n=39)
Peri-impl.
(n=32)
3
2
1
0
−1
−2
−3
−4

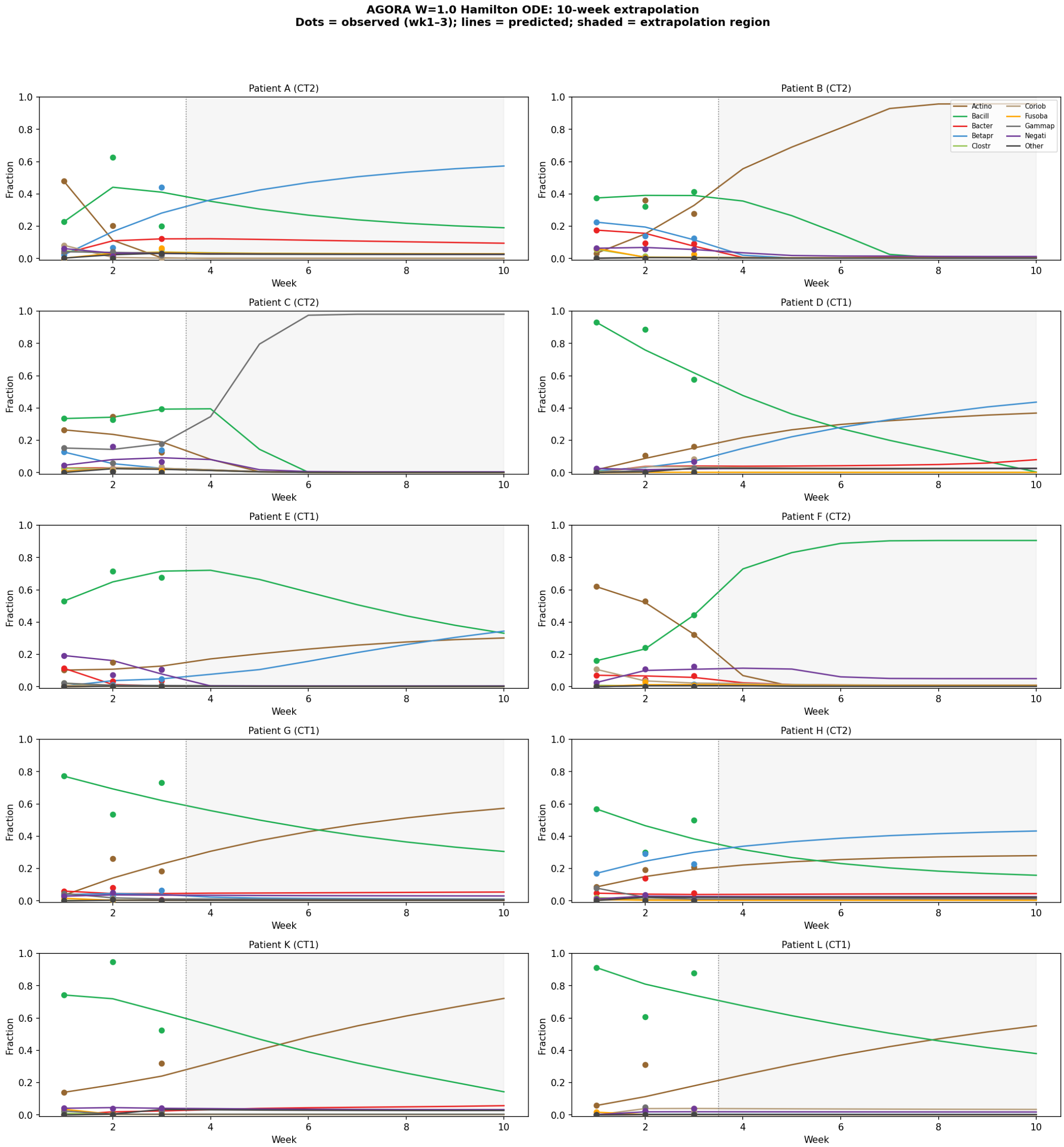


Figure 14: MAP trajectory extrapolation to week 10 (AGORA W= 1.0, all 10 patients). Training window (weeks 1–3) shaded; dots = observed. Solid lines = forward integration with per-patient $\hat{b}$. CT1 patients converge to Actinobacteria/Bacilli-dominated equilibria; CT2 patients to Fusobacteriia/Bacteroidia enrichment. Convergence by weeks 5–7 in most patients. This extrapolation is speculative (no longitudinal data beyond week 3 available for validation) and should be interpreted as a qualitative attractor projection, not a clinical prediction.